

The Geometric Origin of Residue-Class Gap Asymmetry in Sparse Arithmetic Point Processes

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Abstract

We study the conditional gap distribution $P(g \bmod 12 \mid p_n \equiv a)$ for consecutive primes in residue classes modulo 12 and observe a systematic asymmetry: $P(g \equiv 2, 4 \mid a) > P(g \equiv 8, 10 \mid a)$ for all four non-trivial residue classes $a \in \{1, 5, 7, 11\}$.

We identify the residue-class gap asymmetry as a universal consequence of geometric gap statistics rather than a uniquely prime-specific phenomenon. For Bernoulli point processes (discrete Poisson analogues) with constant acceptance probability p , we derive the exact formula $R = (1 - p)^{-6}$, where R is the ratio of gap probabilities. Our data and model analyses suggest that this geometric origin accounts for a substantial portion of the observed asymmetry in primes.

Empirical comparison with three null models reveals an unexpected hierarchy: $R_{\text{Bernoulli}} \approx 1.83 > R_{\text{Prime}} = 1.58 > R_{\text{Cramér}} \approx 1.36$. This indicates that the naive Cramér model (logarithmic acceptance $p(n) = 2/\ln(n)$) *overdamps* the asymmetry through mixing, while real primes show intermediate behavior consistent with sieve-induced correlations.

Our results place primes in an intermediate regime between uncorrelated Bernoulli processes (maximal short-gap bias) and the overdamped logarithmic-density dynamics of the Cramér model, suggesting that arithmetic correlations (sieve structure) partially restore the asymmetry suppressed by logarithmic mixing.

Keywords: Prime gaps, residue classes, Bernoulli process, Cramér model, gap asymmetry, geometric distribution

1 Introduction

1.1 Motivation

The distribution of prime numbers has fascinated mathematicians for millennia. While the Prime Number Theorem establishes the asymptotic density $\pi(x) \sim x/\ln(x)$, the *local* structure of consecutive primes — their gaps $g_n = p_{n+1} - p_n$ — remains rich with patterns and open questions.

Recent investigations [1, 2] have revealed systematic biases in prime gap distributions, notably the Chebyshev bias and prime number races. These phenomena show that primes, despite having equal limiting density in residue classes, exhibit *persistent statistical asymmetries* over finite ranges.

1.2 The Observation

We classify primes $p > 3$ by their residue modulo 12:

$$E : p \equiv 1 \pmod{12}, \quad A : p \equiv 5 \pmod{12}, \quad B : p \equiv 7 \pmod{12}, \quad C : p \equiv 11 \pmod{12}$$

For consecutive primes (p_n, p_{n+1}) with gap $g = p_{n+1} - p_n$, we observe a **systematic asymmetry** in the conditional gap distribution:

$$P(g \equiv 2, 4 \pmod{12} \mid p_n \equiv a) > P(g \equiv 8, 10 \pmod{12} \mid p_n \equiv a) \quad (1)$$

for *all* starting classes $a \in \{E, A, B, C\}$.

1.3 The Question

Is this asymmetry:

- (a) A deep property specific to prime numbers (sieve structure, Hardy-Littlewood correlations)?
- (b) A general phenomenon of sparse point sets (sparsity-induced)?
- (c) Something in between?

1.4 Our Answer

Answer: (c) — but with a surprising twist.

We show that:

- The asymmetry *already exists* in Bernoulli point processes (discrete Poisson analogues) with geometric gap distributions
- It is *not* prime-specific in the strong sense
- But the *magnitude* depends critically on the acceptance model
- Primes occupy an **intermediate regime** between uncorrelated Bernoulli processes and overdamped Cramér dynamics

1.5 Main Results

Theorem 1 (Geometric Asymmetry Formula). *Let $\mathcal{B}$ be a Bernoulli point process (discrete Poisson analogue) on $\mathbb{N}$ with constant acceptance probability p . Define the gap asymmetry ratio:*

$$R_{\mathcal{B}} = \frac{P(g \equiv 2 \text{ or } 4 \pmod{12})}{P(g \equiv 8 \text{ or } 10 \pmod{12})} \quad (2)$$

Then:

$$\boxed{R_{\mathcal{B}} = (1 - p)^{-6}} \quad (3)$$

Theorem 2 (Null Model Hierarchy). *For three sparse point processes up to 10^5 with identical modulo-12 evaluation:*

$$R_{\text{Bernoulli}} \approx 1.83 \quad > \quad R_{\text{Prime}} = 1.58 \quad > \quad R_{\text{Cramér}} \approx 1.36 \quad (4)$$

*This hierarchy is **stable over 10 random seeds** (std. dev. < 0.07).*

1.6 Interpretation

The hierarchy reveals a **two-stage effect**:

1. **Bernoulli** $\rightarrow$ **Cramér** (-28%): Logarithmic acceptance $p(n) = 2/\ln(n)$ creates *mixing* over variable density, strongly damping the geometric short-gap bias.
2. **Cramér** $\rightarrow$ **Prime** ($+16\%$): Primes exhibit intermediate behavior. This is consistent with sieve structure (twins, cousins, sexy primes) enhancing short gaps relative to independent acceptance, though quantitative derivation remains open (Section 7.3).

Conclusion: Primes lie in an intermediate regime between uncorrelated Bernoulli processes (maximal geometric bias) and the overdamped logarithmic-density dynamics of Cramér.

2 The Chirality Observable

2.1 Complete EABC Quadruples

Definition 3. A prime quadruple $Q = (p, q, r, s)$ is **complete** if:

$$\{k(p), k(q), k(r), k(s)\} = \{E, A, B, C\} \quad (5)$$

where $k(p)$ denotes the EABC-class of prime p .

For consecutive complete quadruples $Q_n = (p_n, p_{n+1}, p_{n+2}, p_{n+3})$, we define the **chirality observable**:

$$\chi(Q) = \begin{cases} +1, & \text{if } \text{norm}(\text{sig}(Q)) = EABC \\ -1, & \text{if } \text{norm}(\text{sig}(Q)) = ECBA \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where $\text{norm}(\cdot)$ cyclically rotates the signature to start with E .

2.2 Orientation Bias

The **bias function** is:

$$H(X) = \sum_{Q \leq X} \chi(Q) = N_{\text{EABC}}(X) - N_{\text{ECBA}}(X) \quad (7)$$

Empirical observation (up to $X = 10^5$):

$$h(X) = \frac{H(X)}{N(X)} \approx 0.16 \quad (\text{persistent bias}) \quad (8)$$

2.3 Connection to Gap Asymmetry

The key insight: The orientation bias *follows mechanistically* from gap asymmetry via the deterministic relation:

$$p_{n+1} \equiv p_n + g_n \pmod{12} \quad (9)$$

EABC cycle uses gaps $\{4, 2, 4, 2\} \subset \{2, 4\}$.

ECBA cycle uses gaps $\{10, 8, 10, 8\} \subset \{8, 10\}$.

Therefore:

$$\frac{P(\text{EABC})}{P(\text{ECBA})} \approx \left(\frac{P(g \equiv 2, 4)}{P(g \equiv 8, 10)} \right)^4 \approx R^4 \quad (10)$$

Thus, the **chirality observable is a probe** for the underlying gap asymmetry.

3 Bernoulli Point Process Model: The Geometric Origin

3.1 Setup

Consider a Bernoulli point process (discrete Poisson analogue) on $\mathbb{N}$ where each candidate point is accepted with constant probability p .

Gap distribution: If a point is accepted at position n , the gap G to the next accepted point is **geometrically distributed**:

$$P(G = k) = p(1 - p)^{k-1} = pq^{k-1}, \quad q := 1 - p \quad (11)$$

3.2 Residue Class Distribution

For gaps modulo 12, we have:

$$P(G \equiv r \pmod{12}) = \sum_{\substack{k \geq 1 \\ k \equiv r \pmod{12}}} pq^{k-1} = pq^{r-1} \sum_{j=0}^{\infty} q^{12j} = \frac{pq^{r-1}}{1 - q^{12}} \quad (12)$$

Key observation: The probability *decreases exponentially* with residue class r !

3.3 The Asymmetry Ratio

For our specific residue classes:

$$P(G \equiv 2 \pmod{12}) = \frac{pq^1}{1 - q^{12}} \quad (13)$$

$$P(G \equiv 4 \pmod{12}) = \frac{pq^3}{1 - q^{12}} \quad (14)$$

$$P(G \equiv 8 \pmod{12}) = \frac{pq^7}{1 - q^{12}} \quad (15)$$

$$P(G \equiv 10 \pmod{12}) = \frac{pq^9}{1 - q^{12}} \quad (16)$$

Therefore:

$$R = \frac{P(G \equiv 2) + P(G \equiv 4)}{P(G \equiv 8) + P(G \equiv 10)} \quad (17)$$

$$= \frac{p(q^1 + q^3)/(1 - q^{12})}{p(q^7 + q^9)/(1 - q^{12})} \quad (18)$$

$$= \frac{q^1(1 + q^2)}{q^7(1 + q^2)} \quad (19)$$

$$= q^{-6} \quad (20)$$

$$\boxed{R_{\text{Bernoulli}} = (1 - p)^{-6}} \quad (21)$$

3.4 Interpretation

The asymmetry arises from **short-gap bias**: Geometric distributions favor small gaps, and residue classes $\{2, 4\}$ are *systematically closer to 0* than $\{8, 10\}$ modulo 12.

This is **universal** for any constant- p sparse process with geometric gaps.

3.5 Numerical Prediction

For our empirical Bernoulli processes (mean gap $\approx \ln(50000) \approx 10.82$):

$$p \approx \frac{1}{10.82} \approx 0.092 \quad (22)$$

Predicted:

$$R_{\text{Bernoulli}} = (1 - 0.092)^{-6} \approx 1.85 \quad (23)$$

Observed: $R_{\text{Bernoulli}} \approx 1.83$ (mean over 10 seeds)

Agreement: Excellent!

4 Cramér Model: Logarithmic Damping

4.1 The Cramér Construction

The naive Cramér model accepts each odd candidate n independently with probability:

$$p(n) = \frac{2}{\ln(n)} \quad (24)$$

This gives correct asymptotic density $\sim x/\ln(x)$ but *lacks* sieve structure.

4.2 Why Is It Too Tame?

Key difference: $p(n)$ is *not constant* but *decreases with n* .

For $n \in [5, 100000]$:

- At $n = 1000$: $p \approx 0.289$
- At $n = 50000$: $p \approx 0.185$
- At $n = 100000$: $p \approx 0.173$

Naive prediction: Mean $\bar{p} \approx 0.20$ would give:

$$R_{\text{naive}} = (1 - 0.20)^{-6} \approx 3.81 \quad (25)$$

But we observe: $R_{\text{Cramér}} \approx 1.36$

Discrepancy: Factor of 2.8!

4.3 The Mixing Effect

The gap distribution is *not* purely geometric but a **mixture**:

$$P(G = k) = \int P(G = k \mid \text{start at } n) dP(n) \quad (26)$$

Each window $[n, n + \Delta n]$ has different $p(n)$. This **averaging over variable acceptance rates** strongly damps extremes.

Result: The Cramér model *overdamps* the asymmetry by mixing over variable density.

5 Prime Data: The Empirical Hierarchy

5.1 Methodology

We compare three models up to limit 10^5 :

1. **Bernoulli Process:** Geometric gap distribution with mean $\approx \ln(n)$ (constant p)
2. **Cramér Model:** Independent acceptance $p(n) = 2/\ln(n)$ (variable density)
3. **Real Primes:** Sieve of Eratosthenes

All three undergo *identical* modulo-12 gap analysis: Count transitions and compute:

$$R_{\text{gap}} = \text{geom. mean} \left(\frac{P(g \equiv 2, 4 \mid a)}{P(g \equiv 8, 10 \mid a)} \right) \quad (27)$$

over all four starting classes $a \in \{E, A, B, C\}$.

5.2 Results

Model	Sample Size	R_{gap}	Interpretation
Bernoulli (mean)	~ 6400	1.83 ± 0.06	Geometric baseline
Real Primes	9589	1.58 (fixed)	Intermediate regime
Cramér (mean)	~ 3200	1.36 ± 0.07	Logarithmic overdamping

Table 1: Null model hierarchy over 10 random seeds (Bernoulli and Cramér) at $N = 100,000$.

Hierarchy (consistent over all 10 seeds):

$$\boxed{R_{\text{Bernoulli}} > R_{\text{Prime}} > R_{\text{Cramér}}} \quad (28)$$

Hierarchy (consistent over all 10 seeds):

$$\boxed{R_{\text{Poisson}} > R_{\text{Prime}} > R_{\text{Cramér}}} \quad (29)$$

5.3 Quantitative Breakdown

- **Poisson $\rightarrow$ Cramér:** $\Delta R = -0.47$ (-28%) — Mixing overdamps
- **Cramér $\rightarrow$ Prime:** $\Delta R = +0.22$ ($+16\%$) — Sieve restores
- **Poisson $\rightarrow$ Prime:** $\Delta R = -0.25$ (-14%) — Net regularization

6 Interpretation: Structured Disorder

6.1 The Three Effects

The hierarchy reveals **three distinct mechanisms**:

1. **Short-Gap Bias (Geometric Origin)**
 - Source: Geometric/exponential distributions favor small gaps
 - Formula: $R = (1 - p)^{-6}$ for constant p
 - Observable in: Pure Poisson process

2. Logarithmic Thinning (Mixing Effect)

- Source: $p(n) \sim 1/\ln(n)$ varies with $n \rightarrow$ mixing
- Effect: Dampens asymmetry via averaging
- Observable in: Cramér model

3. Arithmetic Back-Correlation (Sieve Effect)

- Source: Hardy-Littlewood correlations, twin/cousin/sexy prime clustering
- Effect: Enhances short gaps vs. naive independent acceptance
- Observable in: Real primes

6.2 The Equilibrium

Primes sit at an **equilibrium** between:

Geometric short-gap bias	(from sparsity)
Logarithmic mixing	(from density growth)
Sieve back-correlation	(from arithmetic structure)

This creates **structured disorder**: Not maximally random (vs. Poisson), not smoothly damped (vs. Cramér), but a *middle regime* where arithmetic correlations tune the balance.

6.3 Conceptual Spectrum

$$\underbrace{\text{Bernoulli}}_{R \approx 1.83} \xrightarrow{\text{log-mixing}} \underbrace{\text{Cramér}}_{R \approx 1.36} \xrightarrow{\text{sieve}} \underbrace{\text{Primes}}_{R = 1.58} \quad (30)$$

Primes are **more ordered than Bernoulli** (logarithmic growth regularizes) but **more disordered than Cramér** (sieve creates local clustering).

7 Outlook and Open Questions

7.1 Hardy-Littlewood Quantification (Open Question)

The **critical open question**: Quantify which portion of

$$\Delta R := R_{\text{Prime}} - R_{\text{Cramér}} \approx 0.22 \quad (31)$$

can be explained by Hardy-Littlewood k -tuple correlations.

Proposed approach:

1. Compute the expected enhancement of short gaps (2, 4, 6) using:
 - Twin prime constant: $\Pi_2 \approx 0.660$
 - Cousin prime constant: Π_4
 - Sexy prime constant: Π_6
2. Compare this theoretical prediction to the empirical gap ΔR
3. If the Hardy-Littlewood correction quantitatively accounts for ΔR , the causal chain

$$\text{Bernoulli} \xrightarrow{\text{log-mixing}} \text{Cramér} \xrightarrow{\text{sieve}} \text{Prime} \quad (32)$$

would be established.

Current status: This remains an open problem. The present work establishes the geometric baseline and empirical hierarchy, but does not yet close the quantitative gap.

7.2 Modulo-30 Generalization (High Priority)

The formula $R = (1-p)^{-\Delta r}$ should generalize to other moduli. For $m = 30$ (8 residue classes), the predicted hierarchy should persist with adjusted exponents. **Testing this would significantly strengthen the universality claim.**

7.3 Modified Cramér with Sieve

Prediction: A modified Cramér model that *explicitly excludes multiples* of small primes (sieve-Cramér hybrid) should satisfy:

$$R_{\text{Cramér+Sieve}} > R_{\text{Cramér}} \quad (33)$$

approaching R_{Prime} .

7.4 Asymptotic Behavior

Open Question: Does $R(X)$ converge to a constant, or exhibit race-like oscillations as $X \rightarrow \infty$?
Current data show:

$$R_{\text{Prime}}(10^4) = 6.14 \rightarrow R_{\text{Prime}}(5 \times 10^5) = 3.59 \quad (34)$$

This suggests slow convergence, but the asymptotic limit remains open.

8 Conclusion

We identify the observed residue-class gap asymmetry $P(g \equiv 2, 4) > P(g \equiv 8, 10)$ in consecutive primes modulo 12 as arising substantially from the geometric distribution of gaps in sparse point processes, quantified by the exact formula $R = (1 - p)^{-6}$ for Bernoulli point processes with constant acceptance probability.

Our data and model analyses indicate that the naive Cramér model *overdamps* this asymmetry through mixing over variable acceptance probability $p(n) = 2/\ln(n)$, while real primes exhibit intermediate behavior. This places primes between uncorrelated Bernoulli processes (maximal geometric bias) and the overdamped logarithmic-density dynamics of Cramér, consistent with sieve-induced correlations that partially restore short-gap clustering.

Open question: Quantitative derivation of $R_{\text{Prime}} - R_{\text{Cramér}} \approx 0.22$ from Hardy-Littlewood k -tuple correlations would establish the precise contribution of arithmetic structure and close the causal chain from geometric gaps to prime-specific correlations.

Our results reframe the interpretation from “mysterious prime property” to “universal sparse-set phenomenon modulated by arithmetic structure,” opening new avenues for understanding local prime gap patterns through the lens of stochastic point processes.

References

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